

CONDITIONAL NULLS AND POSITIVE CONTROLS FOR TASK-VECTOR SUBSPACE OVERLAP

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ABSTRACT

Task-vector subspace overlap is often interpreted as shared learned structure because it exceeds an isotropic baseline, although linear-layer updates depend on the activations they read. We introduce a conditional randomisation that preserves each update’s spectrum and blockwise occupancy in frozen-base activation geometry while randomising cross-task directional agreement; its expected overlap has a closed form. We pair this null with a positive control that plants shared directions and measures recovery of the delivered signal, exposing consequential under-randomisation in our initial implementation. In complete ViT-B measurements, the corrected activation null reproduces about a quarter of above-isotropic overlap, moving over 20–26% across our calibration and blocking grid at the reported subspace size, and the residual stays positive throughout. Module role moves this far more than architecture does. For shared-start LoRA a regime-matched factor null reproduces about 80%, but a one-line count from the shared rank already gives 82%, so there the admissible row space, not a learned basis, accounts for nearly all observed overlap. These are attribution statements conditional on what each null holds fixed, and they leave pair rankings and downstream merge behaviour unchanged.

1 INTRODUCTION

Independently fine-tuned models built on one pre-trained checkpoint have task vectors whose input-side singular subspaces overlap far above an isotropic random baseline, and a growing body of merging work measures that overlap and builds on it. [Gargiulo et al. \(2025\)](#) report the cross-task inner products of right singular vectors and define an interference score from them. [Zhou et al. \(2026\)](#) define a Right Subspace Overlap statistic over 190 pairs of task vectors and use it as one input to a predictor of merging success. [Qiu et al. \(2026\)](#) construct a null-space projector out of the cumulative update’s right singular vectors, and justify it with a theorem that task vectors approximately align with representation subspaces. In each case the overlap is compared, explicitly or implicitly, against what an unstructured subspace would give.

That comparison is correct for unconditional random subspaces, but it does not answer the conditional attribution question studied here. For a linear map $y = Wh$, the raw minibatch gradient has the outer-product form

$$\nabla_W L_t = \sum_{b \in B_t} g_{t,b} h_{t,b}^\top. \quad (1)$$

Under scalar-step gradient descent without weight decay, the final update’s row space therefore lies in the span of the activations along that specialist’s own training trajectory. This exact statement neither identifies a common frozen-base span nor applies unchanged after coordinatewise optimiser preconditioning: activations drift and optimisers transform gradients. It motivates a weaker, measurable question. Using the frozen base’s activation second moment as a common coordinate system, how much observed overlap is predicted after holding fixed each update’s own occupancy of that geometry? The isotropic value k/d_{in} conditions on none of it.

Prior work motivates both sides of this question. [Cheng et al. \(2025\)](#) relate linear-layer task vectors to their inputs, and [Qiu et al. \(2026\)](#) give an approximate subspace-alignment result; the latter note that the effect of shared initialisation remains insufficiently explored. These results do not establish

frozen-base containment for the checkpoints studied here. Cross-task right-subspace overlap itself is already measured at scale by the work above. What is missing is a stated conditional baseline, an accounting of how much the answer moves under it, and evidence that the baseline randomises what it claims to.

We provide all three. The activation-conditioned null independently acts on each observed update with rotations that preserve its complete singular spectrum and blockwise read-mass profile in the frozen-base activation eigenbasis. Its expectation has a closed form: it is the overlap predicted by the two observed occupancy profiles. The remainder is residual directional agreement conditional on them. This is an operational conditional decomposition; the retained profile is measured after fine-tuning and may itself contain task-learned structure. A generative alternative sampled from C_H is not a valid conditional comparator because it must invent the update–activation relationship and produces a conspicuous level failure in our common-reference stress test.

Applied to the CLIP benchmarks used by this literature, the null reproduces 25.3% and 22.4% of above-isotropic top-8 overlap in the two complete ViT-B measurements. Conditional on the fixed calibration and blocking specification, the remaining excess stays positive throughout the crossed task–layer finite-pool stability interval. Under rank-16 LoRA the answer is mostly the rank constraint. Since $\text{row}(BA) \subseteq \text{row}(A)$, a shared A confines both updates to one 16-dimensional row space, where two independent top-8 subspaces already agree at k/r . An A - and factor-spectrum-conditioned null reproduces 80.3% of the product update’s above-isotropic overlap, against 82.0% from that one-line count. Its attainable plant delivers 37.4–57.0% of the nominal increment and the null recovers 100.7–101.1% of what is delivered, so the construction is calibrated; it simply has little left to explain.

The methodological result is why these estimates, rather than a larger number from our first implementation, are reported. A zero-signal level check accepts the identity map, so it cannot detect under-randomisation. We add a positive control that plants shared directions and measures the signal actually delivered and recovered. It exposed two under-randomisation defects in the activation null and a nuisance-spectrum mismatch in the LoRA null. Correcting the activation null raised delivered-signal recovery from 45.6–51.7% to 99.3–99.6% and moved the full- fine-tuning ViT-B/16 headline from roughly 67% to 25.3%. Surrogates independently drawn from that conditional null also form a relative stress test: the isotropic and participation-ratio-whitened baselines report +0.0171 and +0.0522 apparent excess. This comparison is conditional on our stated reference model and is not an independent intervention.

We make three contributions: a group-conditional decomposition of read-subspace overlap; a positive-control protocol that checks power as well as level and catches consequential defects in our own nulls; and a regime-specific audit of full fine-tuning and LoRA that separates effect-size attribution from pair ranking and downstream merge performance.

2 RELATED WORK

Subspace overlap in model merging. Merging grew from weight averaging (Wortsman et al., 2022; Ilharco et al., 2022; Matena & Raffel, 2022) through task arithmetic (Ilharco et al., 2023; Ortiz-Jimenez et al., 2023) to interference resolved coordinate by coordinate (Yadav et al., 2023; Yu et al., 2024) and then in subspaces (Stoica et al., 2025; Tang et al., 2023; Lee et al., 2025), which is where overlap became a reported quantity. Cheng et al. (2025) relate linear-layer task vectors to their corresponding inputs and Qiu et al. (2026) give an approximate subspace-alignment result. These works motivate activation-aware conditioning; they do not imply exact containment in a frozen-base subspace under feature drift and adaptive optimisation. The overlap measurement is established by Gargiulo et al. (2025), Zhou et al. (2026) and Marczak et al. (2025), none of which states a conditional reference distribution for it. Our contribution is that conditional baseline and a protocol for checking how much coupling it actually randomises.

Activation statistics as machinery. Activation second moments are long-standing, and continual learning has used them longest. Saha et al. (2021) build gradient subspace bases by SVD of stored input activations and Wang et al. (2021) project updates into the null space of the uncentered input feature covariance, the same object as our C_H . Both use it to constrain training. Inside merging the pattern repeats: RegMean (Jin et al., 2023) merges with per-model input Gram matrices computed

on each fine-tuned model’s own data, and two data-free methods estimate the covariance from the parameter differences (Abdel Hameed et al., 2026; Xu et al., 2026). More directly, An et al. (2025) compute pre-linear-layer covariances from each task’s training examples and fine-tuned model to construct a knowledge-aware SVD for pruning task vectors. All target a task-conditioned object, $\mathbb{E}_{x \sim \mathcal{D}_t}[xx^\top]$. We instead choose the frozen base under a recorded calibration mixture as a common, task-agnostic coordinate system; this is part of our estimand rather than a uniquely necessary null. Using $\Delta W^\top \Delta W$ itself as a reference would also be circular: exact inverse-square-root whitening equalises the nonzero singular values and makes a top- k choice non-identifiable rather than supplying an independent calibration geometry. What none supplies is a statement of what an uninformative task vector looks like, and the difference bites: RegMean’s published shrinkage returns $2.3\times$ chance on independent task vectors when repurposed as an evidential baseline (Appendix C).

Null models for similarity claims. Cui et al. (2022) show that representation-similarity metrics such as RSA and CKA (Kornblith et al., 2019; Raghu et al., 2017) are confounded by the population structure of the input data, giving “spuriously high similarity for even completely random neural networks”. Our conditional analysis is the weight-space analogue: there the shared object is population structure in the input distribution, while here the conditioned object is occupancy of a common activation geometry and the measured quantity is task-vector subspace overlap. Ding et al. (2021) make the complementary point for the measures themselves, arguing that a similarity metric must be judged on sensitivity to changes that matter and specificity against changes that do not, which is the pairing §4 imports for nulls. The general form is older. Elsayed & Cunningham (2017) formalise whether population-level structure exceeds what already-known primary features predict, by comparing against surrogate data that preserves those features and is otherwise maximally random. The null of §3 is that construction applied to weights, and §4 is the part of the surrogate-data tradition this literature has not imported: checking what the surrogate actually destroys.

Low-rank adapters. How a shared adapter basis should be read is what §5 bears on, and several methods construct or exploit one (Stoica et al., 2025; Panariello et al., 2025; Zeng et al., 2026). The geometric result has a functional counterpart already known. Zhu et al. (2024) show that in $\Delta W = BA$ the two factors are not interchangeable: fine-tuning B is substantially more effective than fine-tuning A , and a random untrained A performs nearly as well as a trained one. If A need not be trained to work, it is unsurprising that it carries little task-specific information. Ban & Ji (2026) reach the qualitative conclusion directly, reporting that the similarity between A matrices across adapters is attributable to the identical initialisation and not to shared knowledge. That conclusion is theirs. We add the geometric constraint, $\text{row}(\Delta W) \subseteq \text{row}(A)$, and a factor-conditioned null whose reproduced fraction we report with its measured response and scope.

3 A NULL FOR TASK-VECTOR SUBSPACE OVERLAP

3.1 WHAT IS BEING MEASURED, AND AGAINST WHAT

For a pair of task vectors $\Delta W_i, \Delta W_j \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ let $V_i^{(k)}$ hold the top- k right singular vectors of ΔW_i , and write

$$\mathcal{O}(i, j) = \frac{1}{k} \|V_i^{(k)\top} V_j^{(k)}\|_F^2, \quad (2)$$

the mean squared canonical correlation between the two read subspaces. For two independent uniformly random k -subspaces of $\mathbb{R}^{d_{\text{in}}}$ this has expectation k/d_{in} , which we verify numerically in Appendix B; that value is the isotropic null the literature reports against.

The question is not whether $\mathcal{O}$ exceeds k/d_{in} . It does, by an order of magnitude. We ask how much remains after conditioning on specified marginal structure in each update. Throughout, *null-reproduced* means reproduced after holding those stated statistics fixed. It does not identify which process created them: a statistic measured after fine-tuning can itself contain task-learned structure.

3.2 THE ACTIVATION-CONDITIONED NULL

Let $C_H = \mathbb{E}[hh^\top] = V \text{diag}(w) V^\top$ be the input second moment of the *frozen base model* at that layer. This is an operational reference geometry, not the claim that the final update is exactly contained in the frozen-base activation span: activations evolve during fine-tuning, and optimiser

preconditioning can alter the elementary outer-product gradient geometry. Retain the first m_C eigenvectors and partition them into blocks $\{\mathcal{B}_b\}$, with orthogonal projectors P_b ; the unresolved complement is one additional block. Define

$$\mathcal{G} = \{Q \in O(d_{\text{in}}) : QP_b = P_bQ \text{ for every } b\}. \quad (3)$$

For each task independently, draw Q_i from Haar measure on this product of orthogonal groups and set $\widetilde{\Delta W}_i = \Delta W_i Q_i$. This preserves the full singular spectrum because $\widetilde{\Delta W}_i \widetilde{\Delta W}_i^\top = \Delta W_i \Delta W_i^\top$. If $\Pi_i = V_i^{(k)} V_i^{(k)\top}$ and $t_{ib} = \text{tr}(P_b \Pi_i)$, then, away from a singular-value tie at the rank- k boundary, it also preserves every block occupancy t_{ib} and every block energy $\|\Delta W_i P_b\|_F$. It randomises which directions within those allocations two tasks select. It also fixes richer blockwise Gram and principal-angle invariants listed in Appendix B, so the attribution is conditional on more than these summaries.

This construction makes the conditional estimand explicit. Independence and Haar invariance give

$$\mathbb{E}_{\mathcal{N}} \mathcal{O}(\widetilde{\Delta W}_i, \widetilde{\Delta W}_j) = \frac{1}{k} \sum_b \frac{t_{ib} t_{jb}}{\text{rank}(P_b)}. \quad (4)$$

Thus the scalar null-reproduced component is evaluated exactly from the two observed block-occupancy profiles, not estimated from random draws. For this activation null, sampled matrices are used only when actual null task vectors are required—for projector reconstruction, positive controls, and external-reference diagnostics—not for the scalar headline. For a reported pool, the null-reproduced fraction is $f_{\mathcal{N}} = (\overline{\mathcal{O}}_{\mathcal{N}} - \overline{k/d_{\text{in}}}) / (\overline{\mathcal{O}}_{\text{raw}} - \overline{k/d_{\text{in}}})$; averaging over its stated rows precedes the ratio.

The blocks use a fixed operational coarsening. The exact commutant of empirical C_H rotates only its exact multiplicity eigenspaces. Our $\mathcal{G}$ is instead the commutant of a resolution-coarsened, truncated block operator: for a simple retained spectrum its retained action reduces to independent signs, while the deliberately unresolved complement is one full orthogonal block. Finite calibration data motivate merging nearby retained directions as well. We estimate eigenvalue errors from eight disjoint folds and form maximal consecutive runs connected by overlap of adjacent $\hat{\lambda}_j \pm 2\sigma_j$ intervals; Appendix C gives the exact split-sample formula. This is a fixed operational coarsening rule, not a confidence theorem or a unique null. At 1001 images, across all measured modules, both ViT-B encoders give a median 37 blocks over $m_C = 64$ directions; the complement is one full block. Appendix C reports a full $k/m_C/z$ sensitivity grid, the budget dependence, stored-spectrum limit, and rejected fixed-rank alternative; the rule, calibration sample, and measured power are all part of the specification.

Why not a generative null. A generative draw with input covariance proportional to C_H instead invents the update–activation relationship: it concentrates top- k mass toward high-variance activation directions and overlaps far more than the data in a common-reference diagnostic (Appendix N). Coordinate permutation fails for the opposite reason, destroying the occupancy profile despite passing the offset check.

3.3 THE NULL FOR LOW-RANK ADAPTERS

For a LoRA update $\Delta W = \frac{\alpha}{r} B A$ (Hu et al., 2022) the row space lies inside $\text{row}(A)$, and equals it whenever B has full column rank. At initialisation, adapters sharing A_0 therefore have the same admissible r -dimensional row space. This remains exact when A is frozen; when A is trained it need not, so we measure its final overlap rather than assume it. The activation null does not preserve this regime-specific constraint.

Our *factor-conditioned* null keeps each observed A and the singular values of B , and redraws $B \mapsto U \text{diag}(S_B) W^\top$ with independent Haar singular frames. It preserves A and B ’s spectrum, but generally not the product spectrum of BA ; the LoRA estimand therefore has different invariants from the activation null. The product-spectrum- preserving alternative over-randomises A ’s anisotropy, giving +0.2023 offset. Appendix I gives the algebra and alternatives. In the selected null’s positive control, the attainable row-space-constrained plant delivers 37.4–57.0% of its nominal increment and the null recovers 100.7–101.1% of that delivered increment. This establishes measured level and response for that intervention, but does not restore product-spectrum preservation.

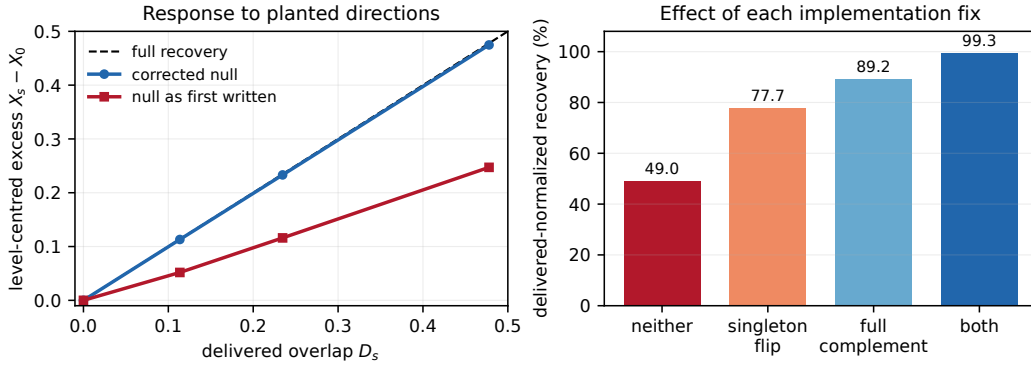


Figure 1: Left: level-centred excess reported against overlap actually delivered by the positive control. The diagonal is full recovery. Right: level-centred recovery at $s/k = 1/4$ with each activation-null correction enabled separately.

3.4 ESTIMATING C_H

For the vision encoders studied here the exact $d_{\text{in}} \times d_{\text{in}}$ matrix is affordable and we use it. Appendix G gives a two-pass estimator for the sizes where it is not, and records why the calibration set is part of the measurement: C_H is an expectation over some input distribution, so we draw it from a mixture across several of the benchmark’s own datasets and record the composition with every covariance.

4 CHECKING A NULL

A null that changes nothing passes a zero-offset check while reproducing all measured overlap. This section pairs that necessary check with a positive control that can expose failure to randomise. We use “level” and “power” descriptively for null offset and planted-signal recovery, not for a rejection threshold, test size, or rejection probability.

4.1 LEVEL IS NOT ENOUGH

The natural check on a null passes a null that does nothing. Build a matched reference pair from an explicitly specified null generator, pass it through the candidate randomisation, and confirm that the reported excess has zero offset. Call this the H_0 check. We report external finite-draw diagnostics separately from the exact scalar decompositions.

Write (A_0, B_0) for a pair from the constructed reference and let $\mathcal{N}_1, \mathcal{N}_2$ be independent candidate-null draws. The H_0 statistic is

$$\mathbb{E}[\mathcal{O}(A_0, B_0) - \mathcal{O}(\mathcal{N}_1(A_0), \mathcal{N}_2(B_0))], \quad (5)$$

and for $\mathcal{N} = \text{id}$ this is exactly zero. A Haar-orbit self-reference is likewise idempotent in distribution, so its population H_0 is exactly zero. The recorded external H_0 instead uses independently activation-generated, spectrum-matched endpoints (Appendix M); because that generator is not the conditional orbit law, its offset mixes reference mismatch, over-destruction, and Monte Carlo error. It is only a stress diagnostic and is never added to the estimand. The identity null can therefore earn a perfect score while randomising little of its target. The asymmetry is not incidental. Eq. 5 measures over-destruction, the error that makes a null report structure where there is none. Under-destruction makes a null report that real coupling as null-reproduced. The check is blind to that failure mode. It is also relative to the constructed reference; it neither establishes a unique data-generating null nor supplies a model-free false-positive rate. The missing diagnostic is a check that can fail under an alternative.

4.2 POWER, MEASURED BY PLANTING

The complement of the H_0 check is a positive control. Take a real task vector A and a partner P_0 drawn from the fitted null, then replace s of the partner’s top- k read directions with directions from

Table 1: Native excess on the same independent-orientation reference pairs, ViT-B/16, $k = 8$, eight fixed modules, and 20 pairs per module (160 total; seed 0). Exact orthogonality subtracts zero; isotropic subtracts k/d ; whitened rows use the participation-ratio floor or all 64 stored directions before subtracting k/d ; the final three rows subtract their own surrogate mean. Thus entries share overlap units but not an estimand.

baseline	reported excess
exact orthogonality (no reference subtracted)	+0.0275
isotropic k/d	+0.0171
whitened at the participation ratio	+0.0522
whitened, all stored directions	+0.0554
generative activation model	−0.8959
coordinate permutation (paired action)	−0.0000

A , projection/QR-complete the remaining directions, and retain P_0 ’s singular values to obtain P_s (exact operator in Appendix L). The intervention introduces forced directional matches on top of the pair’s pre-existing overlap; finite-dimensional cross terms mean that it need not add exactly s/k .

We therefore normalise by what the intervention actually delivers. For candidate null $\mathcal{N}$ define

$$D_s = \mathbb{E}[\mathcal{O}(A, P_s) - \mathcal{O}(A, P_0)],$$

$$X_s^{\mathcal{N}} = \mathbb{E}[\mathcal{O}(A, P_s) - \mathcal{O}(\mathcal{N}_1(A), \mathcal{N}_2(P_s))], \quad R_s^{\mathcal{N}} = \frac{X_s^{\mathcal{N}} - X_0^{\mathcal{N}}}{D_s}. \quad (6)$$

Recovery is thus level-centred and uses delivered rather than nominal overlap. We run this semi-synthetic stress test on real task vectors and the measured calibration covariance. The plant preserves singular values but can change the block-occupancy profile on which the null conditions, so R_s measures sensitivity to this intervention after reconditioning; it is not power against every possible cross-task dependence or evidence about how the conditioned statistics arose.

Figure 1 separates the two arms. The null as first written recovers only 45.6–51.7% of delivered overlap while returning +0.00004 at $s = 0$, which looks well behaved under the offset check; the corrected null recovers 99.3–99.6% and has −0.000623 at $s = 0$. Both arms score the same planted pairs and differ only in the randomisation, which is the combination we want: small reference offset and high response to the planted alternative. The study covers four modules and one family of interventions, and planting can alter conditioned occupancies, so we do not transfer the sub-percent deviation to the 72-module observational estimate.

4.3 WHAT THE POSITIVE CONTROL FOUND

The diagnostic exposed two under-randomisation failures in the activation null: its fixed-rank complement rotation reached only 4–18% of that space, and it skipped singleton sign flips although those directions carry 71% of stored eigenvalue mass. The LoRA generator instead matched only $\|B\|_F$, letting an uncontrolled factor spectrum set even the sign of its offset. The zero-offset check missed the first two and had been declared rather than measured for the third. Appendices I and K give the fixes and ablation.

4.4 A COMMON-REFERENCE STRESS TEST FOR EXISTING BASELINES

A reader’s first instinct is to substitute some other transform for the isotropic comparator, so we show how those transforms behave on pairs whose true overlap is zero by construction. We form semi-synthetic pairs by independently applying our activation-conditioned generator to real updates. None of the cited work proposes its transform as an evidential null; the rows are candidate substitutions, not attributed claims. Residual orientation coupling is absent by construction under this reference, while each endpoint retains its observed spectrum and occupancy profile. Table 1 reports each baseline’s native excess; these are diagnostics on a common input distribution, not one shared estimand, because whitening changes the matrices being scored. Because our generator defines the reference, this experiment neither independently validates our null nor establishes model-free false-positive rates.

The untransformed reference pair’s raw overlap is 0.0275, or 2.6 times isotropic chance; subtracting the isotropic comparator reports +0.0171 excess. Whitening changes the scored matrices and reports still larger native excesses (Table 1). A transform introduced to reduce interference inside a merge therefore need not be calibrated as an evidential baseline.

The mechanism is concentration, and it sharpens with the ridge. $C_H^{-1/2}$ amplifies small eigendirections and drives both endpoints onto the same few of them: at a Tikhonov ridge of $\lambda = 10^{-4}\bar{w}$, 96.5% of the whitened read mass lands on the twenty smallest stored eigendirections, and a chance-overlap pair is reported at 0.7199. RegMean’s published off-diagonal shrinkage $\alpha = 0.9$ returns 0.0931 on the same pair. Truncating the whitening at the top m_{white} directions keeps the reference near chance (0.0394, 0.0405, 0.0422 at $m_{\text{white}} = 8, 32, 64$), which is why we set m_{white} per module from its participation ratio. The failure is therefore not whitening as such but whitening into directions the calibration set does not determine.

The coordinate permutation is the useful counterexample. Here the same permutation is applied to both endpoints, so overlap, and hence its offset, is exactly unchanged; this paired action is a deliberate counterexample rather than the independent action in Eq. 5. It nevertheless destroys occupancy relative to fixed C_H , so §3.2 rejects it. Passing an offset check is not sufficient, which is §4.1’s lesson from the other side.

4.5 THE SAME TEST ON THE LORA NULL

For LoRA, we score only the attainable plant $P_s^* = P_s A^+ A = (\alpha/r) B_s A$, with $B_s = (r/\alpha) P_s A^+$; both raw and null terms use P_s^* (Appendix E). It delivers 37.386%, 49.790%, and 57.043% of the nominal affine increment at $s = 1, 2, 4$. Normalising the level-centred excess by those delivered increments gives 101.117%, 101.078%, and 100.701% recovery. The initial norm-matched generator also has near-100% level-centred response, but its unplanted level is -0.1485 , versus -0.0073 after factor-spectrum matching. The correction therefore calibrates level rather than rescuing low power or an upward bias. This remains one intervention over four modules and six tasks.

5 EXPERIMENTS

Setup. We evaluate CLIP vision encoders fine-tuned on the task-arithmetic benchmark of Ilharco et al. (2023) as extended to twenty tasks by Wang et al. (2024): all 190 pairs over all 72 linear modules for each ViT-B encoder, and 48 q/v modules for ViT-L/14. C_H comes from 1001 images drawn round-robin across seven of the benchmark’s own datasets on the frozen base; a separate rebuilding record stores source counts, fold membership, and hashes. All substantive overlap results use $k = 8$. For whitening comparisons we truncate at the module’s participation ratio m_{white} ; the activation null retains $m_C = 64$ eigendirections and reads its rotation blocks from the eigenvalue standard errors as in §3. The calibration composition, not the calibration draw, is what moves the answer: a deliberately different balanced MNIST/Cars/SVHN mixture gives 21.482% (-3.863 points) while two source-stratified refits at fixed source counts shift it by at most 0.243 points (Appendix C). The residual stays positive in both controls. Brackets are finite-pool stability intervals, the 2.5th and 97.5th percentiles of 50,000 crossed task-layer resampling replicates (seed 20260915). They describe sensitivity to reweighting the observed tasks and layers, not superpopulation coverage, and condition on one checkpoint per task, fixed calibration and blocking, and the exact scalar null (or the pooled 64-draw Monte Carlo null for adapter arms). The external-reference H_0 is a separate stress diagnostic and is never added to an estimate.

5.1 FULL FINE-TUNING: THE CONDITIONAL NULL REPRODUCES ABOUT A QUARTER

In the two complete ViT-B measurements, the activation-conditioned null reproduces 25.3451% ([19.3033, 31.8010]) and 22.3961% ([15.5516, 29.8850]) of above-isotropic top-8 overlap. The partial ViT-L/14 q/v measurement gives 11.8961% ([8.1064, 16.4895]). Conditional on the fixed null specification, the residual is positive under crossed task-layer resampling, but should not automatically be identified with task-derived sharing: it is directional agreement not accounted for by the conditioned statistics. An earlier implementation put its headline reproduced fraction near 67%; the positive control in §4 shows why we reject it, with delivered-signal recovery of only 45.6–51.7% instead of the corrected null’s 99.3–99.6%.

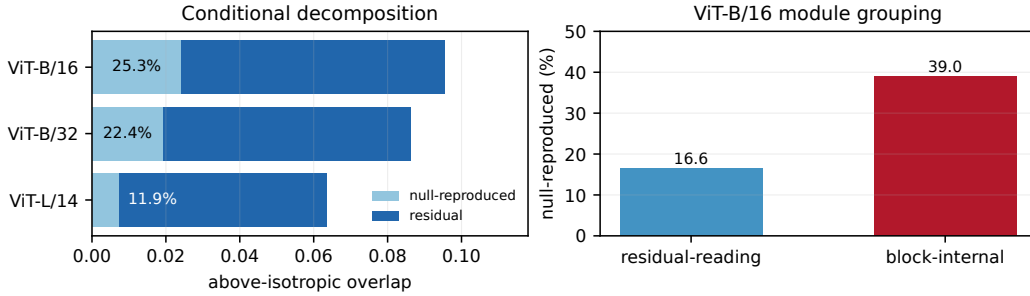


Figure 2: Left: above-chance overlap split into the part the activation null reproduces and the residual for each encoder. Right: an exploratory ViT-B/16 split by whether a module reads the residual stream or a representation built within its block. All fractions use the null expectation without adding the realised H_0 diagnostic.

Module role, not architecture, is where the fraction actually moves, and the pooled headline hides it. Figure 2 shows the grouping: the exact null reproduces 16.6% for ViT-B/16 modules reading the residual stream and 39.0% for modules reading a representation built within the block. Width carries part of that gap, since $fc2$ has four times the input width and supplies 81.9% of the block-internal group’s numerator. Holding width fixed leaves out_proj at 26.2% against 16.1% for attention $q/k/v$, so roughly 10 of the 22 points survive as a role effect. One possible explanation is that task-specific structure accumulates in the residual stream, whereas out_proj and $fc2$ read representations more tightly constrained by the block.

The pooled ratio is a design-weighted blend, and its spread matters for how the headline is read. Across the 72 ViT-B/16 modules the reproduced fraction runs from 0.4% to 65.1% with median 19.1%, and 394 of 13,680 cells have a negative residual. So 25.3% is a property of this module inventory, not of a typical module, and a different inventory would move it.

Table 2: Cross-task read-subspace overlap and its scalar decomposition. Complete encoder pools use all 190 task pairs; the matched eight-task row uses all 28. raw is Eq. 2, iso is k/d , and $null$ is the exact expectation in Eq. 4. Terms are averaged before the ratio, and no Monte Carlo draw enters these scalar results. The last column restricts every pool to q and v , the only module set all three encoders share.

	modules	raw	iso	$null$	$residual$	$reproduced$	q/v only
ViT-B/16	72	0.104574	0.009115	0.033309	0.071265	25.3%	14.7%
ViT-B/32	72	0.095313	0.009115	0.028420	0.066893	22.4%	13.5%
ViT-L/14 q/v	48	0.071402	0.007813	0.015377	0.056024	11.9%	11.9%
matched FT8 q/v	24	0.089930	0.010417	0.023333	0.066597	16.2%	16.2%

Table 2 gives the terms behind these fractions, and its last column shows that most of the apparent spread across encoders is module selection. ViT-L/14 is instrumented on q and v only; restricting the two ViT-B pools to the same two module types moves them to 14.7% and 13.5%, so a 13.5-point gradient becomes a 2.8-point one. We therefore do not read Table 2 as an architecture trend. The fraction is also specification-dependent: across $m_C \in \{16, 32, 64\}$ and $z \in \{1, 2, 3\}$ at $k = 8$ it spans 20.22–25.61% for ViT-B/16, and 13.28–30.37% once k is swept as well, which changes the statistic rather than its estimate (Appendix C).

The residual’s sign is not in question and the resampling does not test it: every one of the 12 layer margins (≥ 0.0334) and all 20 task margins (≥ 0.0282) are strictly positive, so a nonnegative reweighting cannot produce a negative replicate. What the intervals describe is width. For ViT-B/16, ViT-B/32, and ViT-L/14 the crossed-resampling residual intervals are $[+0.04540, +0.10568]$, $[+0.04224, +0.10085]$, and $[+0.03476, +0.08252]$. Within each replicate, independent multinomial endpoint counts C_t and layer counts L_ℓ give row weight $C_a C_b L_\ell$; all module types in a layer stay together, weighted means precede the ratio, and external H_0 diagnostics remain separate.

No single task or layer carries the result. Dropping one task at a time moves the reproduced fraction within 24.685–25.896%, 21.856–22.939%, and 11.310–12.165% for the three encoders; dropping one layer at a time gives 24.359–28.411%, 21.272–26.392%, and 10.955–13.587%. Every residual stays positive in every leave-one-out cell. The layer ranges are the wider of the two, which is what the module-role split above predicts: layers differ in how much of their width reads the residual stream.

5.2 LOW-RANK ADAPTERS: A SHARED-START-COMPATIBLE SIGNATURE

Low-rank adapters make the regime-specific conditioned object explicit. Since $\text{row}(BA) \subseteq \text{row}(A)$, shared A_0 imposes a common admissible row space; this remains exact when A is frozen, while for trained A we measure rather than assume it. Across all 24 targeted modules in the factor audit (Appendix D), learned-from-zero B overlaps at 0.0540–0.0570, well below A 's 0.8759–1.0000 but still about five times chance. The linearized pool, with bit-identical frozen A , has the highest A and product overlap. Almost all of that overlap is the rank constraint, and the null adds little to saying so. The following pool-level estimates use all 24 targeted modules and eight tasks. For LoRA-16 the A -conditioned factor null reproduces 80.3% ([75.7, 85.1]) of above-isotropic overlap, and the linearized pool 76.1% ([72.2, 80.2]). But a shared A confines both updates to one r -dimensional row space, where two independent top- k subspaces already agree at k/r , and the one-line comparator $(k/r - k/d)/(\overline{\text{row}} - k/d)$ gives 82.0% and 74.8% on the same pools. Conditioning on observed A and on B 's spectrum therefore moves the answer by 1.7 and 1.3 points. Under a shared start the admissible row space accounts for nearly all observed product-subspace overlap, and the factor null confirms that rather than discovering it. Its attainable plant delivers 37.4/49.8/57.0% of the nominal increment with level-centred recovery 101.1/101.1/100.7% at $s = 1/2/4$, so the construction is calibrated.

Full fine-tuning over the same 24 modules gives 16.2% ([9.2, 27.5]). We do not read the gap against 80.3% as one contrast: they are ratios with different denominators, computed under nulls that condition on different objects.

A regime-mismatched null changes the attribution. The activation null instead leaves 99.9% of the same LoRA pool's above-chance overlap as residual, versus 19.7% under the observed- A -preserving factor null (+0.5966 versus +0.1176). It answers a different question because it does not hold observed A fixed; transferring a null across regimes therefore changes the inferred effect size substantially.

5.3 CONSEQUENCES: ATTRIBUTION RATHER THAN A NEW MERGE

We rebuild the null-space projector of Qiu et al. (2026) from real and independently randomised task vectors (Appendix J). Observed-null basis overlap is 0.8750 versus 0.0208 chance for LoRA and 0.2594 versus 0.1667 for full fine-tuning. The corresponding basis mass in the top activation subspace is 0.1356 and 0.2612 (chance 0.0833); none of these projector statistics has a dedicated positive control. This does not test projector utility. A separate cross-term-only compression heuristic does not beat the cumulative-update basis at any rank budget, losing on 6 to 8 of the eight tasks from $r = 32$ upward and tying within noise at $r = 8$ (Appendix F). Calibration is also nearly rank-preserving over task pairs (Spearman 0.994–0.995); we do not test a retrained mergeability predictor. The contribution is therefore inferential: a large effect or useful basis does not by itself establish the mechanism invoked to explain it.

6 CONCLUSION

Cross-task top-8 sharing remains positive after conditioning in every fully fine-tuned CLIP pool, but its attribution changes and its size is not a property of the architecture. The null reproduces about a quarter in the complete ViT-B runs, module role moves that far more than encoder scale does, and the shared-start LoRA value is mostly the rank constraint. These quantities split agreement under stated conditions; they do not locate its origin in the base or the shared start.

Three things follow for work that measures subspace structure. State the baseline as an invariant, since writing down what is held fixed makes the estimand checkable. Treat an offset check as necessary

and not sufficient, since a null that destroys too little still returns zero on a reference it generated. And keep a null in the regime it was built for.

REPRODUCIBILITY STATEMENT

Appendix H records task pools, calibration data, numerical settings, and checks; the supplementary artifact provides row-level results, executable analysis, and manifests.

USE OF LARGE LANGUAGE MODELS

Large language models were used for typo checking, grammar correction, and wording suggestions. The authors take full responsibility for the paper’s content.

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A LIMITATIONS AND FUTURE WORK

All quantitative conclusions concern CLIP vision encoders, with ViT-L/14 restricted to $\mathbf{q}$ and $\mathbf{v}$. The decomposition is conditional on a stated reference, and the reproduced fraction is composition-sensitive, so it reports how much overlap survives a stated conditioning rather than where the agreement originates. Each task contributes one checkpoint, so training-run variation is unmeasured, and the stability intervals are finite-pool rather than superpopulation statements. The LoRA result covers rank 16 under one shared-start protocol. We leave broader calibration compositions, language models, and a controlled shared-versus-independent- A_0 experiment to future work.

B INVARIANTS OF THE OVERLAP STATISTIC

Every numerical property below is checked directly in the analysis; these checks should accompany the released measurement code rather than be inferred from library defaults.

Beyond the scalar occupancies and energies in §3.2, the group action fixes $\Delta W_i P_b \Delta W_i^\top$ and the eigenvalues of $V_i^{(k)\top} P_b V_i^{(k)}$ for every block. The former contains output-side block Gram geometry and the latter its principal-angle spectrum. A looser null conditioning only on t_{ib} could therefore give a different attribution.

Why Eq. 4 is exact. Haar averaging within block b gives $\mathbb{E}[Q_i^\top \Pi_i Q_i] = \sum_b (t_{ib}/d_b) P_b$, where $d_b = \text{rank}(P_b)$. For independent task-wise draws, linearity and $P_b P_c = \mathbf{1}\{b = c\} P_b$ give $\mathbb{E} \text{tr}(Q_i^\top \Pi_i Q_i Q_j^\top \Pi_j Q_j) = \sum_b t_{ib} t_{jb} / d_b$; division by k yields Eq. 4.

- **Normalisation.** $\mathcal{O}(A, A) = 1$ to 10^{-6} , and two independent uniformly random k -subspaces average 0.03995 against $k/d = 0.04000$ over 300 draws. We confirm the same identity at $(d, k) \in \{(200, 8), (200, 32), (400, 8), (3584, 8), (100, 10)\}$.

- **Silent failure modes.** Three are worth naming because each returns a plausible number rather than an error: an SVD orientation swap puts U on the input side and reverses every conclusion; the SVD of a zero matrix returns a fixed arbitrary basis, so untargeted adapter modules score $\mathcal{O} = 1.0000$ and inflate any unguarded pool average; and suffix-matching module paths against a full CLIP checkpoint matches `text_model.*` as readily as `vision_model.*`. The released code refuses all three rather than resolving them silently.

C CALIBRATION CHOICES FOR WHITENING AND ACTIVATION BLOCKS

Block-error rule. Partition the calibration set into eight disjoint folds and compute the ordered covariance eigenvalues $\lambda_j^{(f)}$ independently on each fold. With their mean $\bar{\lambda}_j^F$, the implementation uses the split-sample standard error $\sigma_j = [\sum_{f=1}^8 (\lambda_j^{(f)} - \bar{\lambda}_j^F)^2 / (8 \cdot 7)]^{1/2}$. The blocks are the transitive consecutive components induced by overlap of adjacent $\hat{\lambda}_j \pm 2\sigma_j$ intervals, where $\hat{\lambda}_j$ is the eigenvalue from the full calibration pool. This is an operational coarsening, not a confidence interval for eigenspace identifiability.

Finite calibration refits. Holding the seven source counts and extraction procedure fixed, two source-stratified bootstrap refits (seeds 20260916 and 20260917) give full ViT-B/16 exact reproduced fractions 25.102% and 25.302%, versus the 25.345% headline, for a maximum 0.243-percentage-point shift. This is the observed range of two refits, not a calibration confidence interval, and does not test a different source composition.

Composition stress control. Separately, we sample without replacement from deterministic test splits to balance 334 MNIST, 334 Stanford Cars, and 333 SVHN images (seed 20260919). This deliberately different calibration gives exact ViT-B/16 null 0.029621, residual 0.074953, and reproduced fraction 21.482%, or 3.863 percentage points below the seven-source headline. Holding this covariance fixed, the crossed finite-pool stability intervals are $[16.8975, 26.2690]\%$ for the fraction and $[+0.048606, +0.109015]$ for the residual. The median participation ratio is 12.691, with 71/72 modules below 64, so both composition and spectral effective rank have changed. This is a stress control, not an estimate of calibration uncertainty or an interval over possible compositions.

Truncation. §4.4 reports how aggressive whitening manufactures overlap and how truncation removes it. We set m_{white} per module from that module’s participation ratio $(\sum_i w_i)^2 / \sum_i w_i^2$, used here as an operational effective-rank truncation. In our calibration refits, directions beyond this cutoff were unstable; the participation ratio is not an identifiability threshold. The ratio must be computed from the full spectrum: taken over only the m_C retained eigenvalues it is bounded above by m_C by Cauchy–Schwarz, which understates it and makes any gate comparing the two a tautology.

Why a block count was rejected. §3 sets the rotation blocks from what the calibration data can resolve. We record here what the alternative looked like, because the alternative is what a reader would reach for first.

Fixing a block count and complement rotation rank leaves two choices that no invariant selects, and the answer moves with them: sweeping the rotation rank of the complement from 32 to 512 moved the reproduced fraction from 48% to 23% under the null as it then stood. Subtracting that null’s offset on a constructed reference did not stabilise it. The resolution-based construction eliminates those two outcome-sensitive choices, since the complement is a single unresolved block and is rotated in full. It does not eliminate design choices: the uncertainty multiplier, fold-SE scheme, calibration sample, and stored dimension remain part of the null specification.

Block structure is a function of the calibration budget. Blocks are the groups of eigendirections that our operational interval rule treats as unresolved. They tend to sharpen as the calibration set grows: estimated eigenvalue errors contract, so fewer adjacent intervals overlap and blocks split. This behavior does not make interval overlap a confidence theorem for eigenspace identifiability. The first three columns below are measured on nested subsets of one image pool. At each budget we partition images into eight disjoint folds and compute the eight foldwise eigenvalue replicates used by the

split-sample rule in §3. The remainder scales the measured standard errors, the pool having stopped at 1001.

calibration images	measured			extrapolated		
	224	500	1001	2000	5000	∞
ViT-B/16	16	25	37	46	55	64
ViT-B/32	15	21	37	44	55	64

This is a resource-resolution trade-off that the method makes visible. Too few images and the blocks are coarse, the null rotates across gaps that are in fact resolved, and its constructed-reference offset grows. As the budget grows, the rule tends to split blocks. Under a simple population spectrum and fixed $m_C = 64$, its retained-region limit is $\{\pm 1\}^{64}$ while the unresolved complement remains $O(d_{\text{in}} - 64)$. Thus the limit is not the identity: signs randomise off-diagonal terms within the retained region, and the complement is fully rotated. It has no power against agreement expressed only as shared retained-axis occupancy, while retaining power against sign-sensitive or complement orientation alignment. The ∞ column extrapolates the retained block count only; it is not a measured power result.

Block counts at 224 images vary across draws of the calibration set, resolving 13 and 11 blocks on an independent draw where the nested subset resolves 16 and 15. We therefore fix the budget at 1001, where the two architectures agree at a median of 37 blocks, and report it as part of the specification.

Where the blocked region ends. The activation-null stored dimension m_C divides the space the null treats two ways: inside $\text{span}(V)$ it rotates within resolvable blocks, outside it rotates in full. We sweep $k \in \{4, 8, 16, 32\}$, $m_C \in \{16, 32, 64\}$, and uncertainty multiplier $z \in \{1, 2, 3\}$ on each reported pool:

pool	$k = 8$, all (m_C, z)	full (k, m_C, z) grid
ViT-B/16	20.22–25.61%	13.28–30.37%
ViT-B/32	17.94–22.55%	13.31–28.51%
ViT-L/14 q/v	5.84–12.00%	5.36–19.01%

At fixed (k, m_C) , the largest observed change from varying z is 0.754 percentage points; most of the fixed- k range comes from m_C . This is variation in the conditional estimand—storing more directions changes what the null holds fixed—rather than numerical instability of its exact estimator. We pre-specify the headline at $m_C = 64, z = 2$; m_C remains distinct from the per-module whitening truncation m_{white} above.

The calibration must use the observed spectrum. The reference offset is not a property of the null alone; it depends strongly on the spectrum of the task vectors the null is calibrated against. Substituting a synthetic exponential spectrum for the observed one raises the realised reference offset at eight blocks from +0.0041 to +0.0254, a factor of six. This sensitivity is another reason not to transfer that diagnostic into the estimand. The reason is that real task vectors have rapidly decaying spectra, which makes the top- k subspace well determined and the block rotation effective, whereas a flat spectrum leaves that subspace ill-determined and the block structure dominates. A null calibrated against a spectrum the data does not have calibrates the wrong quantity.

D LORA FACTOR AUDIT

At the $k = 8$ boundary, the relative singular gap $(s_8 - s_9)/s_8$ is above 10^{-3} for every observed A and BA cell in both eight-task, 24-module adapter pools. Across 24,576 factor-null BA redraws, only four fall below 10^{-3} and none below 10^{-4} ; the median gap is about 0.099 in each arm. Thus exact or numerical cutoff ties do not drive the aggregate factor-null result, although the finite- k choice remains part of the estimand.

Table 3: ViT-B/16, eight tasks, 28 pairs, all 24 targeted attention modules; all entries use top-8 singular subspaces. B is initialised at zero and is therefore entirely learned; A is not. `lin-LoRA` is the partial linearization of Tang et al. (2024), which leaves A bit-identical across tasks.

	A right	B left	ΔW right
LoRA-16	0.8759	0.0540	0.6075
lin-LoRA	1.0000	0.0570	0.6649
isotropic	0.0104	0.0104	0.0104

E POWER OF THE LORA NULL

Table 4: A rank-16 LoRA stress test, ViT-B/16, four attention modules, six tasks. The fitted factor null has pre-planting overlap $\mathcal{O}_0 = 0.4825$. From an unconstrained plant P_s we form the attainable $P_s^* = P_s A^+ A = (\alpha/r) B_s A$, where $B_s = (r/\alpha) P_s A^+$. Both raw and null terms score P_s^* . Delivery divides Δ_{raw} by $(s/k)(1 - \mathcal{O}_0)$; recovery is level-centred as in Eq. 6.

	planted	raw	delivery	initial X_s	corrected X_s	recovery
0/8	0.4825	—	—	-0.1485	-0.0073	—
1/8	0.5067	37.386%	—	-0.1243	+0.0172	101.117%
2/8	0.5469	49.790%	—	-0.0840	+0.0579	101.078%
4/8	0.6301	57.043%	—	-0.0009	+0.1414	100.701%

The initial norm-matched generator also has approximately 100% level-centred response, but a large $s = 0$ error. Spectrum matching changes that level from -0.1485 to -0.0073 while retaining near-full response. The correction therefore fixes level; it is not evidence that the initial null had low power or upward bias. Values just above 100% reflect finite draws.

F MERGED ACCURACY BY RANK BUDGET

Table 5: Merged accuracy averaged over the eight tasks, ViT-B/32, 2000 test images per task, $\lambda = 0.3$, update compressed to rank r . Zero-shot is 48.3% and uncompressed task arithmetic is 70.3%, both within a point of the published values for this benchmark. *tasks* counts how many of the eight the τ basis wins outright. Per-task binomial standard errors are about 1.1 points, so the mean gap at $r = 8$ is within noise.

rank r	8	16	32	64	128	256
τ basis	56.6	60.2	64.0	66.4	68.4	69.7
cross-term basis	56.0	59.1	61.7	64.0	65.6	66.4
tasks won by τ	4/8	5/8	7/8	6/8	7/8	8/8

The separate cross-term-only heuristic uses $\sum_{t \neq s} \Delta W_t^\top \Delta W_s = \tau^\top \tau - \sum_t \Delta W_t^\top \Delta W_t$. It does not improve compressed-merge accuracy. The τ basis has the higher mean at every budget and the gap widens with r , from 0.6 points at $r = 8$ to 3.2 at $r = 256$. The per-task counts locate where that is decisive: at $r = 8$ the two bases split the tasks 4/8 and the mean gap sits inside one standard error, while from $r = 32$ upward τ wins 6 to 8 of the eight and at $r = 256$ all of them. Concentrating empirical cross-task coupling is therefore insufficient for better compressed-merge accuracy at useful budgets, and indistinguishable from the cumulative-update basis at the smallest one.

Prompt protocol. Zero-shot heads average the benchmark’s per-dataset prompt ensembles, which range from one template for the digit tasks to 18 for `resisc45`. This matters more than it looks: an earlier run of this table used a single three-template generic ensemble and put both the zero-shot and the uncompressed task-arithmetic accuracy about ten points

low, which would have made the comparison above a statement about a non-standard baseline. That run is retained as `results/v7_merge_b32_accuracy.json`; the table reports `results/v8_merge_b32_accuracy.json`.

Calibration is close to rank-preserving. Across the 190 ViT-B/32 pairs, raw and calibrated overlap have Spearman 0.995 and share 17 of the top 20 pairs; ViT-B/16 gives 0.994 and shares 19 of the top 20. Thus an ordering-only consumer receives nearly the same ordering on these pools, although we do not retrain the mergeability predictor of Zhou et al. (2026).

G ESTIMATING C_H

For the encoders studied here the exact matrix is affordable (594 MB across all 72 modules of ViT-B/32) and is what we use. At widths where it is not, a two-pass estimator works: a Frequent Directions sketch locates the subspace, and a second exact pass inside it repairs the downward eigenvalue bias that the sketch introduces, which matters because whitening applies $C_H^{-1/2}$. We do not exercise that path in this paper.

H REPRODUCIBILITY RECORD

Checkpoints and task pools. The frozen bases are `openai/clip-vit-base-patch16`, `openai/clip-vit-base-patch32`, and `openai/clip-vit-large-patch14`. Full-fine-tuned specialists follow `tanganke/<architecture>.<task>`. The 20 task labels are `cifar10`, `cifar100`, `dtd`, `emnist_letters`, `eurosat`, `fashion_mnist`, `fer2013`, `food101`, `gtsrb`, `kmnist`, `mnist`, `oxford-iiit-pet`, `oxford_flowers102`, `pcam`, `rendered-ss2`, `resisc45`, `stanford-cars`, `stl10`, `sun397`, and `svhn`. The matched eight-task comparison uses `dtd`, `eurosat`, `gtsrb`, `mnist`, `resisc45`, `stanford-cars`, `sun397`, and `svhn`; its adapter repositories append `_lora-16` or `_l-lora-16`. Rank and scale are read from each `adapter_config.json`, and untargeted modules are dropped rather than treated as zero subspaces.

Calibration and numerical settings. The rebuilding record contains 143 images from each of `cifar10`, `dtd`, `eurosat`, `gtsrb`, `resisc45`, `stl10`, and `sun397`; fold sizes are 126, 125, 125, 125, 125, 125, 125, 125. Images are the first valid examples encountered in the streaming training split of each `tanganke/<source>` dataset, converted to RGB. Extraction uses each model’s `CLIPImageProcessor`, all sequence tokens (class and patch), exact uncentred $\sum hh^T/n$ accumulation in float32, $m_C = 64$, and CPU execution. The extractor-input basename/size hash common to the three covariance manifests has prefix `8f88002f7557`; the separate logical image/fold record has identifier-list-hash prefix `73797f525986`. Full digests are in the artifact.

Randomisation and verification. The activation-null scalar results are the exact expectation in Eq. 4; they use no random draws. On the matched full-fine-tuning pool, a sampled check gives null 0.02330 versus exact 0.02333. External-reference diagnostics use finite draws, the two factor-null adapter arms use four independent 16-draw batches (seeds 0, 1000003, 2000003, and 3000017), pooled row-wise to 64 draws, and projector rebuilds use six. The batch-based pooled Monte Carlo standard errors (sample SD divided by $\sqrt{4}$, three degrees of freedom) are 0.0067 and 0.0054 percentage points for LoRA-16 and linearized LoRA-16; pooling changes their seed-0 16-draw fractions by -0.0109 and -0.0093 points. The common-reference baseline uses 20 pairs for each of eight fixed modules at seed 0. The crossed stability intervals draw 50,000 replicates at seed 20260915. Independent multinomial endpoint counts C_t and layer counts L_ℓ give each row weight $C_a C_b L_\ell$; all module types within a selected layer remain grouped, and each replicate takes the ratio after weighted averaging. The intervals hold the calibration draw, fold partition, block rule, and m_C fixed, and adapter arms additionally hold their pooled 64-draw realised Monte Carlo null values fixed. `analysis/check_paper_numbers.py` recomputes point estimates from row-level CSVs, and `analysis/overlap_bootstrap.py` reproduces the 50,000 crossed-resampling stability intervals. The artifact pins immutable revisions for all bases, specialists, adapters, and calibration datasets, together with a locked runtime environment. Large model weights, covariance tensors, and calibration images are not bundled.

I SPECIFICATION OF THE RANDOMISATION, AND THREE WAYS IT FAILS

The complement was rotated inside a fixed frame. ΔW splits into a part inside $\text{span}(V)$ and a residual R whose rows lie in the complement. The first version drew a random 128-dimensional frame of that complement and rotated R inside it, leaving the remainder bit-identical between two task vectors. The complement is $d_{\text{in}} - m_C$ dimensional, so this covered 18% of it at $d_{\text{in}} = 768$ and 4% at $d_{\text{in}} = 3072$, while about three quarters of the real read mass lives there. The fix needs no square Haar matrix: R has rank at most $\min(d_{\text{out}}, d_{\text{in}} - m_C)$, so a Haar map on the complement carries its right singular vectors to a uniform frame of that rank, which one thin SVD and one QR deliver exactly.

Singleton blocks were skipped. Blocks of size one were left alone on the grounds that a one-dimensional space cannot be rotated. Its orthogonal group is $\{+1, -1\}$, whose Haar measure is a fair coin. Independent flips preserve diagonal coordinate masses and randomise off-diagonal cross-coordinate signs, which cancel in expectation, while leaving $\Delta W \Delta W^\top$ and hence the spectrum untouched. On ViT-B/16 the singletons carry 71% of the stored eigenvalue mass, so skipping them retained substantial coupling.

The LoRA generator matched a norm where a factor spectrum was required. The factor-conditioned null of §3.3 preserves A and the singular values of B ; it does not generally preserve the product spectrum of BA . Its first implementation redrew B as an i.i.d. Gaussian at matched Frobenius norm, leaving even the factor spectrum uncontrolled: a Gaussian is nearly Marchenko–Pastur where trained B decays. On fitted-reference pools with shared A and independent B , the flat-spectrum generator reports +0.33 when B has no decay and −0.16 when it does. Drawing B at its observed singular values holds the reference offset within +0.008 across the same sweep. This removes the observed factor-spectrum mismatch; it is not a claim of exact product-spectrum preservation. Indeed the nonzero squared singular values of the redraw are the eigenvalues of $\text{diag}(S_B)W^\top(AA^\top)W\text{diag}(S_B)$, which depend on W unless $AA^\top$ is isotropic. A uniform product-spectrum-preserving rotation inside $\text{row}(A)$ fails differently: it discards this anisotropy and has constructed-reference offset +0.2023. Rotating only to BQ retains a learned singular frame and changes the product singular values by 19.7%.

J PROJECTOR REBUILD

Table 6: Projector-basis overlap after independent draws from each fitted conditional null, and the observed basis’s mass in the top-64 activation subspace. ViT-B/16, eight tasks, $r_p = 128$ over 12 full-fine-tuning modules and $r_p = 16$ over 24 adapter modules.

pool	observed-null	chance	in activation	chance
full fine-tuning	0.2594	0.1667	0.2612	0.0833
LoRA-16	0.8750	0.0208	0.1356	0.0833

Qiu et al. (2026) filter an incoming task vector with $P = I - VV^\top$, where V contains the top- r_p right singular vectors of the cumulative update τ . We build this basis once from the observations and once after independently applying the regime-specific null to every task. The activation null preserves full-update spectra and occupancies; the LoRA null preserves its factor-level invariants. The second column reports overlap of the resulting bases, while the fourth is a separate concentration statistic. Neither identifies how that structure arose, tests accuracy, or contradicts the gains reported for the projector. A basis reproducible under a conditional null cannot, without further evidence, identify the cross-task mechanism used to justify it. The LoRA value has no projector-level positive control; the factor-null overlap stress test covers only one different, narrow intervention.

K WHICH CORRECTION DID THE WORK

The two corrections to the activation null can be switched independently. Scoring the same planted pairs at $s = 2$ with each combination separates them.

full complement	singleton flip	level-centred recovery
		49.0%
	✓	77.7%
✓		89.2%
✓	✓	99.3%

Each arm is centred by its own $s = 0$ level and divided by the same delivered increment as in Eq. 6. The singleton flip alone raises recovery from 49.0% to 77.7%, the full complement alone to 89.2%, and both to 99.3%; the effects overlap. The singleton flip is not a token gesture. On ViT-B/16 the singletons hold 71% of the stored eigenvalue mass, and leaving them fixed exempted most of $\text{span}(V)$.

L PLANTED-STRUCTURE RECOVERY, BY LEVEL

Planting operator. Let $P_0 = U_0 \text{diag}(\sigma_1, \dots, \sigma_\rho) V_0^\top$ be a thin SVD with descending singular values, and let $S = V_A[:, 1:s]$ contain the source task’s first s right singular vectors. We set the first s columns of $\hat{V}$ to S . In order, we project $V_0[:, s+1:\rho]$ onto $S^\perp$ and apply QR/Gram–Schmidt against all columns already accepted; any algebraic or numerical rank deficiency is filled by a Haar frame in the remaining complement. We then set $P_s = U_0 \text{diag}(\sigma_1, \dots, \sigma_\rho) \hat{V}^\top$. Hence $\hat{V}^\top \hat{V} = I$ and every singular value and left singular vector of P_0 is retained. Away from a tie at the relevant singular-value boundary, the first s leading right directions match the source exactly. The same planted matrices are scored by every candidate null.

Table 7: Positive-control response on ViT-B/16, four modules, six tasks. `initial` is the first activation null and `corrected` is §3.2. Both score the same planted pairs. `delivered` is $(\text{raw}(s) - \text{raw}(0))/[(s/k)(1 - \mathcal{O}_0)]$. Recovery is the level-centred excess increment divided by the delivered overlap increment, Eq. 6; it is therefore comparable across planting levels and is not the uncentred excess divided by s/k .

planted	raw	delivered	initial recovery	corrected recovery
0/8	0.02771	—	$X_0 = +0.00004$	$X_0 = -0.000623$
1/8	0.14129	93.45%	45.6%	99.6%
2/8	0.26235	96.53%	49.5%	99.4%
4/8	0.50557	98.30%	51.7%	99.3%

M OVERLAP DECOMPOSITION, IN FULL

Table 2 in §5.1 gives the aggregate terms. For the exploratory ViT-B/16 role grouping in Figure 2, the same unadjusted ratio is 16.5568% for residual-reading modules (`q`, `k`, `v`, `fc1`) and 38.9609% for block-internal readers (`out_proj`, `fc2`).

On the matched FT8 pool, a sampled-matrix validation gives mean null 0.02330, versus the exact 0.02333. Sampled matrices remain necessary for projector and positive-control experiments.

Table 8: External-reference H_0 stress diagnostics. These activation-generated references are not the exact Haar self-orbit, whose population $H_0 = 0$ by identity. The finite-draw offsets can mix reference mismatch and over-destruction and are never added to an estimate.

pool and candidate action	realised external H_0
matched FT8, activation null	−0.00040
LoRA-16, activation null	−0.00037
lin-LoRA, activation null	−0.00037

N THE GENERATIVE NULL AND WHY IT FAILS

The obvious construction is generative: draw a random update whose input-side covariance is proportional to the base activation covariance, adding an isotropic draw in the unresolved complement. It does not work because it must invent the relationship between the trained update and C_H . Independent draws then concentrate on the same high-variance activation directions even when observed updates have a different occupancy profile. On the explicitly scoped eight-task, eight-module common-reference diagnostic in Table 1, this complement-aware generator reports native raw-minus-null excess -0.8959 , whereas the observed-profile-preserving null reports -0.0007 . These are baseline-specific stress levels, not estimates of the headline conditional fraction. An earlier exploratory generator omitted the unresolved complement entirely; we discard its quantitative output rather than using a known-defective construction. The comparison locates the requirement precisely: a conditional null here must retain the observed update–activation occupancy rather than generate that relationship from an unverified model.